



**Verified Carbon
Standard**

VERIFICATION REPORT FOR UMUT-II HYDROPOWER PLANT, TURKEY PROJECT



Document Prepared By Earthood Services Private Ltd.

Contact Information

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Prepared By	Earthood Services Private Limited
Contact	1203-1205, 12th Floor, Tower B, Emaar Digital Greens, Sector - 61, Gurugram, Haryana, India – 122011 Telephone: +91 124 4204599, +91 9871747771 Website: www.earthood.in Email: info@earthood.in
Approved By	Dr. Kaviraj Singh – Managing Director
Work Carried Out By	Team Leader and Technical Expert (TA 1.2): Ranjan Singh Trainee Team Leader/ Verifier (New): Akanksha Sengupta Verifier (Old): Abhishek Koul Meth Expert: Shifali Guleria Local Expert: Kubra Agriman Technical Reviewer (New): Jinesh Amlani TA Expert (TA 1.2) to TR (New): Jinesh Amlani Technical Reviewer (Old): Shreya Garg TA Expert (TA 1.2) to TR (Old): Shreya Garg

Summary:

Scope of verification

Earthood Services Private Limited (hereafter referred to as ESPL) has been contracted by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. to conduct the verification of the project - “UMUT-II Hydropower Plant, Turkey Project”, VCS ID 1011 with regard to the relevant requirements of VCS programme guidelines and standard (VCS standard version 4.5/07/, & VCS program guide version 4.4/06/). Relevant requirements of the UNFCCC for CDM project activities, as well as criteria for consistent project operations, monitoring and reporting has been applied for verification.

The monitoring period covers under this verification is from 21-February-2014 to 20- February -2020 (both days included).

The verification includes confirming the implementation of the monitoring plan of the registered PD and the application of the monitoring methodology as per ACM0002, Version 12.1.0 “Grid connected electricity generation from renewable sources”.

The project involves construction of a large-scale Hydro Power project which is a run-of-river hydro power plant near the Town of Akkus on Karakus River in Ordu Province of Turkey. The total installed capacity of the technology is 25.305 MWm / 24.450 MWe. The project activity is developed by RHG Enertürk Enerji Üretim ve Ticaret A.Ş.

The project activity involves electricity generation by the Hydro power supplying the generated electricity to the national grid of Türkiye. The operational start date of the project activity is 21/02/2014 i.e. the day the PA started generating electricity which was checked from the provisional acceptance document/25/. Thus, the project aims to displace electricity produced by fossil fuel power plants harnessing kinetic energy of run off river. Therefore, the project reduces greenhouse gas emissions and thereby contributes to sustainable development.

A risk-based approach has been followed to perform this verification. In the course of verification, 03 Corrective Action request (CARs), 01 Forward Action request (FARs), and 05 Clarification request (CLs) were raised and successfully closed.

The review of the project design documentation, monitoring report and additional documents related to baseline and monitoring methodology; the subsequent background investigation, telephonic interviews and stakeholders have provided ESPL with sufficient evidence to validate the fulfillment of the stated criteria.

ESPL confirms that the project is implemented in accordance with the registered PD. The monitoring system is in place and the emission reductions are calculated without material misstatements. Our opinion relates to the project's GHG emissions, and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the information seen and evaluated we confirm that the emission reductions from the project activity "UMUT-II Hydropower Plant, Turkey" in Türkiye during the period 21/02/2014 to 20/02/2020 (including both days) amount to 115,715 tonnes of CO₂e.

Summary of the verification opinion

Earthood has performed the verification of the VCS project activity "UMUT-II Hydropower Plant, Türkiye". The assessment team has confirmed that:

- The PA is in accordance with all relevant host country criteria (Turkey) and VCS rules and requirements/26//6/7/.
- The PA is in accordance with all conditions of the latest version of applied methodology ACM0002, Version 12.1 "Grid connected electricity generation from renewable sources"/8/
- The monitoring plan is transparent and adequate.
- The installed equipment essential for measuring parameters required for calculating emission reductions are properly maintained.
- The monitoring system is in place and functional. The project has generated GHG emission reductions.

All the information has been consistently applied in the VCS-PD/1/.

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1 INTRODUCTION

1.1 Objective

RHG Enertürk Enerji Üretim ve Ticaret A.Ş. has contracted Earthood Services Private Limited (ESPL) to perform second VCS Verification of the 'UMUT-II Hydropower Plant, Türkiye' in Türkiye (hereafter called project). This project has already been registered as a VCS project (VCS ID 1011). The objective of this verification is a thorough and independent assessment of registered project activities against the applicable VCS requirement by the VVB. The verification process shall determine whether the project activity complies with the requirements of latest VCS guidelines, applicability conditions of the selected methodology, relevant host country regulations and guidance issued by the VCS Board.

1.2 Scope and Criteria

The scope of verification is to assess the claims and assumptions made in the VCS monitoring report (MR) against the VCS criteria, including but not limited to, VCS standard, applied methodology and other relevant rules and requirements established for VCS project activities.

The Verification is not meant to provide any consulting towards the project participants. However, stated requests for clarification and/or correction actions request may have provided inputs for improvement of the project design. There are no material errors, overestimation of ER, omission or misstatement. Verification team conducted remote audit to verify metering/monitoring arrangement. The verification team has reviewed all the documents like provisional acceptance document/25, technical specification, Calibration certificates /39,12/, meter readings/18/, grievance register/14/ etc.

1.3 Level of Assurance

☒ Reasonable Level of assurance

☐ Limited Level of assurance

Earthood's verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the verification by obtaining evidence, conducting the on-site visit and other information and explanations that Earthood considers necessary to give reasonable assurance that reported estimated GHG emission reductions are fairly stated. During the current verification the VVB conducted the on-site visit for the project activity as discussed under section 2.2 of this report and no substantial changes were observed thereby meeting the reasonable level of assurance. In our opinion, the estimated GHG emissions reductions were calculated correctly based on the approved baseline and monitoring methodology ACM0002, Version 12.1/8/, and the VCS standard, Version 4.5/7/. The evidence used to achieve a reasonable level of assurance is specified in section 2.3 and 2.4 of this report.

1.4 Summary Description of the Project

The project activity "UMUT-II Hydropower Plant, Turkey Project" involves a run-of-river hydro power plant including 3 turbines of the Vertical axis Francis type are installed with 8.435 MWm / 8.150 MWe capacity each totaling to a capacity of 23.305 MWm / 24.450 MWe MWe. The hydro power plant is located on the Karakus stream in Ordu province of Turkiye. The project harnesses the renewable energy to generate and connected electricity to the National electivity grid, and therefore replacing the electricity that would have been generated through the carbon intensive fossil-fuel power plant or by addition of new units in the future. Hence, reducing the probable GHG emissions produced. The implementation of the project has been verified through on-site visit observation/28/ and through desk review/appendix 1/.

The project start date for UMUT-II HEPP is considered as 21/02/2014 which is also the first time the electricity generated was supplied to the national electricity grid replacing electricity generated by fossil fuels and reducing GHG emissions. This was verified by the verification team from the VCS PD/1/ and cross-checked from the provisional acceptance document/25/. The project activity applies the large scale CDM methodology ACM0002 version 12.1: "Grid connected electricity generation from renewable sources"/8/.

On-site visit/28/ was conducted on 04/10/2023 and it was confirmed that the current monitoring period which is the first verification is from 21/02/2014 to 20/02/2020 (including both days). The project activity in the current monitoring period replaces by delivering 210,015.91 MWh electricity to the national grid leading to overall 115,715 tCO₂e emission reductions throughout the current monitoring period and 19,286 tCO₂e annual emission reductions.

2 VERIFICATION PROCESS

The registered VCS project is undergoing first verification under VCS, the approach adopted to ensure the quality of emission reductions is described in the following sections.

2.1 Method and Criteria

The verification approach consists of two phases:

In the first phase, ESPL completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;
- Means of verifying reported data; and
- Compilation of the verification Report.

At the end of this phase, ESPL produced a Verification Checklist which, based on the risk assessment of the parameters and data collection and handling processes for each of those parameters, describes the verification approach and the sampling plan.

In the second phase using the Verification checklist, ESPL verified the implementation of the monitoring plan and the data presented in the VCS MR/2/ for the period in question. This involved telephonic interviews of project proponent representative's and a desk review of the Monitoring Report. This verification report describes the findings of this assessment. The findings raised during the assessment can be found in appendix III of this report.

Duration of Verification

Verification Contract	05-06-2023
Kick Off Meeting	15-07-2023
Desk Review Initiated	03-08-2023
On Site Visit	03-10-2023
Findings Raised	17-11-2023
Final Verification Report	Version 1.0, dated 29/02/2014

2.2 Document Review

The verification is performed primarily as a document review of the documents submitted at various stages of assessments. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the documents (Project Description/1/ & Monitoring Report/2/) and information from sources other than those used, if available, and also conducts independent background investigations. Earthood conducted a desk review as under;

- a) A review of the data and information presented to verify their completeness;
- b) A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures;
- c) An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On site audit was conducted for project activity on 03/10/2023. During the site visit, audit team visited the facility and interviewed PP representatives/28/ to discuss various aspects of the project activity such as implementation of the project activity, monitoring plan, operation and maintenance, details on calibration, local stakeholder consultation, grievance mechanism etc. Names of the people audited, as can be confirmed from VVB attendance audit/28/, are given below:

No.	Interviewee			Date and Location	Topics Covered	Team member
	Last name	First Name	Affiliation			
1.	Yurtoglu	Deniz	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık ve Ticaret A.Ş.	03/10/2023 Akkus Town, Turkiye	Operation and Maintenance of Facility, Monitoring Practices, Data collection, Logbooks, QA/QC procedures, calibration procedures and details, Grievance Mechanism, major shutdown in plants, data recording and management, training imparted	Ranjan Singh, Kubra Agriman
2.	Karabekiroğlu	Bahadır	RHG Enertürk Elektrik & Karbon Ticareti			
3.	Saffet	Arsalan	RHG Enertürk Elektrik & Karbon Ticareti			
4.	Turk	Cemil	Local Stakeholder/ (Candarli Neighbourhood)		- Following topics were discussed about the Local Stakeholder Consultation: - Meeting Invitation - Grievance mechanism - Topics covered during the LSC Meeting - The effects of the project activity	
5.	Turk	Remzi	Local Stakeholder/ Mukhtar			
6.	Turk	Necaattiin	Local Stakeholder/ (Candarli Neighbourhood)			

2.4 Site Visits

An on-site audit was conducted for the project activity on 03/10/2023.

As per para 4.1.12 of the VCS Standard, v4.5/7/ “A site visit that includes a visit to facilities and/or project areas shall be conducted at validation. Such a site visit shall be conduct at verification under the following circumstances:

1. The first verification of the project after validation.
2. Verification of project baseline reassessments; and
3. Verifications that assess a project description deviation where the deviation impacts the applicability of the methodology, additionality or the appropriateness of the baseline scenario.”

The scope of this assessment is the 1st verification of the project activity, hence VVB team physically visited the site for assessment.

Representatives of PP along with on ground staff and local stakeholders were interviewed about various aspects of project activity like implementation, monitoring procedure, QA/QC procedure, calibration procedure, training of the site personals. The VVB toured the entire site and captured photographic evidence of key technical and social mechanisms in place. Audit team also inspected the functionality and efficiency of all equipment at the station to ensure that on ground working corroborated with the facts stated in MR by PP. This was deemed sufficient to confirm the plant operation and other aspects of the project. Our approach was in accordance with para 4.1.2 of the VCS standard v4.4 “The validation/verification body shall select samples of data and information to be validated or verified to provide a reasonable level of assurance and to meet the materiality requirements of the specific project.” Sufficient amount of data and information was collected during current verification which were included in appendix I of this report, to provide a reasonable level of assurance. The representatives of PA were also interviewed, details of which has been included in section 2.3 of the report.

On- site audit Date – 03/10/2023			
S.no	Activity Performed On-site	Site Location	Team Member

1.	Assessment team checked the implementation of the project. Baseline emission, Emission reduction calculation, technical description of the project and monitoring. Assessment team also checked that weather the monitoring plan as described in PD is actually practiced on -site.	Akkus Town, Karakus River, Ordu, Turkiye <table><tr><th colspan="2">Power House</th></tr><tr><td>Longitude</td><td>37° 01' 37" E</td></tr><tr><td>Latitude</td><td>40° 42' 48" N</td></tr><tr><th colspan="2">Diversion Weir</th></tr><tr><td>Longitude</td><td>37° 04' 0.07" E</td></tr><tr><td>Latitude</td><td>40° 42' 4.90" N</td></tr></table>	Power House		Longitude	37° 01' 37" E	Latitude	40° 42' 48" N	Diversion Weir		Longitude	37° 04' 0.07" E	Latitude	40° 42' 4.90" N	Ranjan Singh, Kubra Agriman
Power House															
Longitude	37° 01' 37" E														
Latitude	40° 42' 48" N														
Diversion Weir															
Longitude	37° 04' 0.07" E														
Latitude	40° 42' 4.90" N														

The on-site observations were corroborated with other evidence and supportive referred during desk review. The technical details/16/ and metering monitoring arrangement verified through onsite photographs/28/ of name plates and calibration certificates/13,39/. All the documents were cross-checked to ensure conservative estimation of emission reduction. To verify the information, various documents like commissioning certificates/25/, Generation License/33/ technical specifications/19/, Plant records, EPIAS screenshots/11/, etc. were also assessed. The complete list of documents is given in Appendix I.

2.5 Resolution of Findings

The objective of this step is to identify, discuss and conclude on the issues which are material and that are found during the assessment (Monitoring, implementation, and operations) of the project activity which could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of the emission reductions. The findings may be of the following types: CAR – Corrective action Request, CL – Clarification Request, FAR – Forward Action Request.

Based on the above assessment it can be concluded that the assessment team has verified sufficient data, to reduce audit risk to an acceptably low level as a requisite to achieve reasonable level of assurance for the current monitoring period

CAR (Corrective Action Request) is raised if one of the following occurs:

Non-compliance with the monitoring plan, the methodology or the standardized baseline are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient.

- Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants.
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions.
- Change to the key sustainable development indicators.
- Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable requirements have been met. All CARs and CLs raised by the Earthood during verification shall be resolved prior to submitting a request for issuance.

FAR (Forward Action Request) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

The Objective of this verification phase was to resolve the requests for corrective actions, clarification and any other outstanding issues from verification which need to be assessed for Earthood's positive conclusion on the Monitoring report. The corrective action requests and clarification requests raised by the VVB (Earthood) were successfully resolved by the PP (RHG Enertürk Enerji Üretim ve Ticaret A.Ş.) to complete the verification process.

A total of 05 CLs, 03 CAR and 01 FAR were raised in the current verification. All the findings that are raised and communicated to project participant during the verification are included under Appendix III. The section also includes the response, if provided, by the project participants and an assessment by the verification team if it was closed out or otherwise.

2.5.1 Forward Action Requests

This is the first verification of the Project Activity and no FAR was raised from validation/05/.

However, 1 FAR has been raised during current verification pertaining to the Start date and the duration of crediting period.

As per registered PD the first crediting period was expected to begin on 01/12/2011 for a duration of 10 years till 30/11/2021. However, given that the actual generation of electricity only began on 21/02/2014, as confirmed by the VVB from Provisional Acceptance documents, the actual start date of the PA

A FAR has been raised for the subsequent VVB to ensure that the VCS Registry is updated to reflect the same, post 1st verification.

2.6 Eligibility for Validation Activities

Not applicable since there was no such undertaking for the given project.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

Project is registered under VCS only (VCS ID-1011)/1/. It is confirmed from the declaration/17/ provided by the project proponent “RHG Enertürk Enerji Üretim ve Ticaret A.Ş” that the net GHG emission reductions generated by the project activity will not be used for compliance with any other emissions trading program or to meet binding limits on GHG emissions for the same monitoring period.

The project activity is not a part of any other GHG Program. The VVB has independently assessed the CDM project webpage and also, checked other registries like GS and GCC for the registration of the project activity.

The credit issuance records for the PA (VCS - 1011) are as verified from VCS webpage, accessed from - <https://registry.verra.org/app/projectDetail/VCS/1011>, is summarized as:

Sr. No.	Monitoring Period	Verifying VVB
1.	21/02/2014 to 20/02/2020 (1st First monitoring period)	Earthood Services Private Limited

Hence, the VVB concludes that the PA is registered under VCS and is only claiming the ERs through VCS program and is not registered in any other GHG programs.

3.2 Methodology Deviations

No methodology deviations were found for this monitoring period.

However, clarification regarding calculation of Electricity Generation has been provided by PP during verification, which was not included in the registered monitored plan as stated in the PD. The VVB assessment for the same is expounded in section 3.3 and 4.5 of this report.

3.3 Project Description Deviations

The verification found that the deviations in the project are detailed in the Generation License issued by EMRA. The initial generation licensee/29/, "Nisan Elektromekanik Enerji Sanayi Ve Ticaret A.S," transferred shares to Boydak Enerji Üretim Ticaret Anonim Şirketi, which later changed its name to "RHG Enertürk Enerji Üretim ve Ticaret A.Ş.," who is the current license holder. VCS was notified of the company name change on 17/01/2022 and has been confirmed via copy of communication/30/.

The VVB has assessed the amendments to the Generation License (License Number: EU/1590-1/1156) /29/, which states that 24/01/2014 and 03/09/2021 which document the change in name from "Nisan Elektromekanik Enerji Sanayi Ve Ticaret A.S" and "Boydak Enerji Üretim Ticaret Anonim" respectively.

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A distinct metering method for gross electricity generation and consumption was implemented at UMUT-II HEPP. Instead of separate metering, UMUT I HEPP, UMUT II HEPP and UMUT III HEPP, are connected to the same grid point, share one meter, providing only combined gross electricity production data. The project was expected to begin on 01/10/2012 The start date for a however is 21/02/2014, as confirmed from the Provincial Acceptance Document /25/.

The project owner asserts using a constant ratio between the capacities of the three plants to calculate UMUT-II HEPP's individual gross electricity production from the combined data.

EPIAŞ records values for the three plants combined is considered for calculation. Additionally, EPIAŞ and TEİAŞ systems provide data on gross electricity generation and internal electricity consumption, with transmission losses already deducted from the former. A constant multiplier of 0.5787 is used for calculations related to UMUT-II HEPP's electricity generation and internal consumption.

The constant multiplier of 0.5787 is calculated as the ratio of project capacity of UMUT-II to the combined capacity of UMUT HEPP (UMUT I + UMUT II+UMUT III). This calculation has been verified by the VVB and the capacities of all three plants have been found to corroborate with both the Generation License (License Number: EU/1590-1/1156)/29/ and Provisional Acceptance Document (/25/

The modifications include the shift from PMUM to EPIAŞ occurring in monitoring operations, affecting the data source for electricity generation and consumption figures.

Throughout the Monitoring Report, PMUM has been replaced by EPIAŞ to reflect this change. The Emission Reduction Excel Spreadsheet specifies the data sources for each month during the monitoring period/2/.

In addition to this there has been a change in the start date of the project activity. As per the registered PD/01/ the 'expected' start date was to be 01/10/2012. However, as confirmed by VVB from the provincial acceptance document/25/, the actual start date i.e. the date from which the PA started electricity generation, is 21/02/2014. The date on registry will be updated post submission of verification opinion by VVB. A FAR has been raised for the same.

In line with the provided guidelines, these deviations were reviewed by the verification team and found to be thoroughly documented. The project's conformity with VCS rules and justifications for these deviations have been outlined. An overall assessment determines the validity of these deviations in the context of the project's adherence to VCS regulations.

VVB confirms that the forementioned project deviations do not impact the appropriateness of the baseline scenario, additionality, or applicability of the methodology .

3.4 Grouped Project

The project is not part of a grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project activity is the implementation of hydro power plant with a total capacity of 25.305 MWm / 24.450 MWe located near the Akkus Town, on the Karakus River in Ordu Province of Türkiye. The same has been confirmed from the project KML file/31/ and geocoordinates notes during the on-site audit/28/. The purpose of this project activity is to generate electricity through renewable energy sources and then this generated electricity is exported to the Turkish electricity grid. Since, this is the first verification of the current crediting period, the commissioning, capacity, project implementation, grid connectivity, and technical specifications have been once checked at validation.

In the Sub-Station two energy meters are installed—one serving as the primary meter and the other functioning as a backup meter. Both meters are the property of TEİAŞ. TEİAŞ conducts meter readings through their Remote Reading System and communicates this information to the power plant management system. Additionally, backup meters are positioned within the Substation to oversee energy production values. TEİAŞ enables remote access to the main meters and monitors the generated energy quantities every 15 minutes. The verification of energy production occurs through the comparison of values between the primary and backup energy meters.

The start date of the project activity is 21/02/2014, which is the date the PA started generating electricity, the same has been verified through the provisional acceptance/25/ for PA. The first crediting period of the project activity is from 21/02/2014 to 20/02/2024, with 10 years period and can be renewed twice. The current monitoring period is the first verification of the first crediting period starting from 21/02/2014 to 20/02/2020 (including both days).

In agreement with national legislation no full-EIA was conducted for the Project. The VVB has confirmed the same from the Government of Türkiye's Official Gazette No:26939/32/and the EIA exempt certificate of the Provincial Ministry of Environment issued for Umut 1-3 HEPP/37/.

The project activity will reduce the GHG emissions generated by the current generation energy mix in Turkish national grid, which is dominated by carbon intensive fossil fuel-based grid connected power plants. The power generated through the project activity is then sold to the national electricity grid, which has been verified through the EPIAS screenshots/11/ shared by the PP and same has been dictated in section 3.1 of MR.

TEİAŞ is responsible for overseeing and ensuring the quality and accuracy of the meters. The calibration of these meters is exclusively managed by TEİAŞ, thereby eliminating the need for internal audits by power plant personnel. The calibration process complies with national regulations and adheres to the Measuring Instruments Directive. As per the Regulation on Metering and Metering Devices, meter calibrations remain valid for 10 years. During meter replacements, which occur on the same day, there is a period of zero generation. These replacements are conducted in adherence to the revised System Use Agreement/34/ between the Project Proponent (PP) and TEİAŞ, and testing, performed every two years, is the sole responsibility of TEİAŞ concerning both the tests and the meters.

Based on the above auditing measures it was concluded by the verification team that there was no material discrepancies between the actual monitoring system, and the monitoring plan set out in the VCS PD/1/ and the applied methodology/8/, This has been verified from the review of information available in MR/2/ and cross-checked against joint VCS PD/1/ and the applied methodology "ACM0002 version 12.1"/8/.

The technical specifications of the project activity equipment's were verified during the On-site observations/28/ and manufacturer technical specifications document/16/ provided by PP and all the equipment's installed on site were found to be consistent with the details mentioned under monitoring report/2/.

The verification team conducted interviews with the plant representatives to confirm that the generation data of the inverters is accurately reported by cross checking the data from the SCADA monitoring system and can therefore confirm that the project is fully functional.

A declaration certificate/17/ shared by the PP, has clearly stated that the project activity has not received or sought any other form of environmental credit or is doing so during current verification. A declaration certificate/17/ shared by the PP, has clearly stated that the GHG emissions reduction or removals generated by the project are not the part of emission trading program or any other mechanism that includes GHG allowance trading. The project activity is a grid-connected Hydro power plant which is contributing to SD by reducing the GHG emission, that would have been emitted from the project baseline.

The project activity contribution to the sustainable development for the current monitoring period is as described below:

SDG Indicator	Contribution due to project activity
SDG 13.2.2	115,715 tonnes of CO ₂ e per annum reduced due to electricity generation from hydro power plant as calculated based on EPIAS records/11/.
SDG 7.2.1	210,015.91 MWh of renewable energy generated due to project activity as verified from EPIAS records/11/.
SDG 8.5	Employment generation of 12 persons and training were provided as verified from employment and training records/24/.

Based on the above, it can be concluded that the project has been implemented as described in registered PD/01/ and no discrepancies has been found in this regard.

The assessment team concludes the following:

1. There are no material discrepancies between project implementation and the project description provided in the registered PD/1/.
2. The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling, and analysing the monitored data and parameters) is appropriate.
3. There is no material discrepancies between the actual monitoring system, and the monitoring plan set out in the project description and the applied methodology/7/.
4. It has been confirmed by the assessment team that PP is not claiming emission reductions for the same monitoring period in both the programmes. PP has confirmed and provided a no double counting declaration for the same/9/.
5. The project has not received or sought any other form of environmental credit or has become eligible to do so since validation or previous verification.
6. The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading/9/.
7. The project activity is complying with indicators for sustainable development as discussed in detail under section 1.11 of MR/2/.

Based on the verified information and the above conclusions, the assessment team can conclude that the project has been implemented as described in the project description.

CL 03, CL 04, CL 05, and CAR 02 were raised and resolved.

4.2 Safeguards

4.2.1 No Net Harm

Since the project is a grid connected renewable energy power plant which involves generation of electricity through hydro power, it does not have any negative environmental and social impacts as was checked during the review of the PD/1/ and the grievance register/17/.

In compliance with national laws, a complete Environmental Impact Assessment (EIA) was not carried out for the Project. This was confirmed by the VVB through the Government of Türkiye's Official Gazette No: 26939/32/ and the EIA exemption certificate issued by the Provincial Ministry of Environment for Umut 1-3 HEPP. /37/

However, the project proponent conducted an EIA, and the report// has been duly assessed and no negative impacts on the environment have been identified. Further the waste water/40/ records have been assessed

4.2.2 Local Stakeholder Consultation

The Local stakeholder consultation has been conducted before commissioning of the PA on 21/05/2009 in Akkus town. A secondary stakeholder feedback was conducted in the same region in 2010. Apart from these, various visits were undertaken to local communities to understand their views on the PA. In addition to consulting with the local populace, the Turkish Ministry of Environment and Forestry was actively engaged in the consultation process. Same was confirmed during the interviews /28/ for on-going stakeholder communication.

PP has maintained a Continuous Input/ Grievance Expression Process book at the site of the HEPP, maintained by the chief of the HEPP, as confirmed from site observation. Local stakeholders can lodge their grievances anytime in the register over the operational lifetime of the project activity.

During the current monitoring period, no grievances were received as has been reported under section 4.2.2 of the MR/2/. This was confirmed by assessing the grievance register// on site which had no entries pertaining to the concerned monitoring period. Further, it was noted that during on site interview with stakeholders//, none of them reported any grievance or negative impact arising from the project activity. aThus, the verification team is of the opinion that the ongoing stakeholder's mechanism is adequate and appropriate.

4.3 AFOLU-Specific Safeguards

This a non AFLOU project activity. Hence, this section is not applicable.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

The calculation of the emission reductions is found to be correct. The details of the reported and the verified values for all parameters are listed in section 4.5 of this report.

The parameter EG_{facility,y} is directly sourced from monthly settlement notification issued by Turkish Electricity Transmission Company. The PP has provided the complete set of data for all the monitored parameters in the ER spreadsheet/4/. This data has been verified as described in section 4.5 below. The formulae & method used to calculate the baseline emissions, project emissions and leakage are appropriate and in line with the approved methodology ACM0002 version 12.1/8/.

As per ER excel spreadsheet/4/ submitted by the PP, the net emission reductions for the current monitoring period were verified as 115,715 tCO₂e for the current monitoring period.

The calculation method and formulae used in calculating baseline emission is complying to the methodology used i.e., ACM0002 version 12.1/8/. Since project activity is a Hydro power project, leakage emission and project emission has been considered as zero because the power density is more than 10 W/m².

According to the assessment team, monitoring for the relevant period was conducted in accordance with the protocols outlined in the registered documents, and the emission reductions that resulted were both quantifiable and conservative.

Means of verification	The verification team evaluated the information flow in the MR/2/ (from data generation, calculation, and reporting of parameters, including values). The verification team examined whether the calculations of baseline GHG emissions, project GHG emissions, and leakage GHG emissions were performed in accordance with the formulae and methods described in the monitoring plan of the PD/1/.
Finding	CL 02, CAR 1 and CAR 03 were raised and resolved
Conclusion	As the project activity claims emission reductions from electricity generation, the baseline emissions for a given year are computed in accordance with “Tool to calculate the emission factor for an electricity system (v02.1.0)”/14/ of approved methodology ACM0002 v12.1.0/8/. The verification team certifies that the proper procedures and formulas for computing baseline emissions have been used. The computations used reasonable assumptions, emission factors and default values.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The ER's generated for the project activity is based on the electricity generation from the solar power plants. The PP has shared all the relevant evidence to substantiate the data related to the monitored parameter. The list of all the documents that were reviewed for the current verification has been listed down in appendix I of this report. The values of the parameter, electricity imported and exported has been listed down in the appendix I of this report. The values of the parameter, electricity imported and exported has been checked by the verification team through the monthly EPIAS Screenshots/11/ and meter reading reports/18/ to verify the values reported in the ER Sheet/4/ and were found to be appropriate. Since the EPIAS screenshots are prepared and issued by the TEİAŞ, they are found to be an authentic and reliable source and the Meter reading reports/18/ are being issued by the PP. The PP has taken the values reported in the EPIAS Screenshots for the calculation of emission reductions since EPIAS Screenshots is an official governmental document used to document the amount of electricity transferred to the grid. The approach is conservative after assessment and was found to be appropriate.

The monitoring of the project activity is found to be in accordance with the monitoring plan set out in the VCS PD/1/ and the approved methodology ACM0002 version 12.1/8/. The verification of ex-ante and monitored parameter has been discussed below:

Assessment of Data and Parameter:

The assessment of following ex-ante parameters mentioned in the PD/1/ has been carried out as per the method described in the table below:

Ex-ante parameter	Assessment
Official emissions from power generation, National GHG Inventory of Turkiye submitted to UNFCCC	PP has applied the value in line with table 10 mentioned in Appendix of MR and these values have been sourced from the 2007-2009 official emissions data submitted to UNFCCC for power generation facilities in Turkiye. It's an ex-ante value fixed at the time of validation. The value mentioned was checked and found to be consistent with the VCS PD/1/.
GEN_{j,y}, The gross electricity generation by fuel type j in year y (2005-2009) by the system (MWh)	PP has applied the value in line with the calculation method used in 'table 9. Gross and Net electricity Generation (GWh) in Tukiye of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The value mentioned was checked and found to be consistent with the VCS PD/1/.

System Net Electricity Generation , The difference between the total quantity of electricity generated by power plants/units and the auxiliary electricity consumption of power plants/units (MWh)	PP has taken the value as per the calculation method used in Table 9 'Gross and Net Electricity Generation (GWh) in Tukiye of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/is found to be consistent with the VCS PD/1/.
Net Delivery Ratio , The ratio of the System Net Electricity Generation to the System Gross Electricity Generation in year y (%)	PP has taken the value as per the calculation method used in Table 9 'Gross and Net Electricity Generation (GWh) in Tukiye of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/ is found to be consistent with the VCS PD/1/.
GEN_y , the system net electricity delivered in year y by the total of all power plants in the grid, excluding lc-mr sources (MWh)	PP has taken the value as per the calculation method used in Table 9 'Gross and Net Electricity Generation (GWh) in Tukiye of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/ is found to be consistent with the VCS PD/1/.
EF_{CO2,i,y} , CO2 emission factor of fossil fuel type i in year y (tCO ₂ /TJ)	PP has taken the value as per the lower limits of the 95% confidence interval stated in the "2006 IPCC Guidelines for National Greenhouse Gas Inventories", Volume 2, Chapter 1 (energy) Table 1.4/36/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/ is found to be consistent with the VCS PD/1/.
Electricity Import_y , Net electricity imports delivered to the grid in year y (GWh)	PP has taken the value as per the calculation method used in Table 9 'Gross and Net Electricity Generation (GWh) in Tukiye of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/ is found to be consistent with the VCS PD/1/.
η_{m,y} , Average net energy conversion efficiency of power unit m in year y (%)	PP has taken the value as per the calculation method used in Table 11. Calculation of the BM

	emission factor of the registered PD/1/. It's an ex-ante value fixed at the time of validation. The source of the value mentioned in the MR/2/ is found to be consistent with the VCS PD/1/.
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The assessment of following monitored parameters mentioned in the PD/01/ and the MR/3/ has been carried out as per the method described in the table below:

Parameter	Net electricity supplied by the project activity in year y EG PJ, facility, y (MWh)												
Means of verification	PP has applied the value of the electricity generation and was checked by the verification team with the EPIAS records/11/ and meter reading records/18/. The calculation for the parameter was checked from the ER Sheet/4/. The value is monitored by the PP during verification. The value mentioned was checked and found to be consistent with the MR/2/. The verification team cross-checked the parameter is mentioned in the VCS PD/1/.												
	Criteria/Requirements	Assessment/Observation											
	Measuring /Reading /Recording frequency	Continuous measurement and at least monthly recording.											
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The reporting frequency is in line with the monitoring plan as outlined in the registered VCS PD/01/ and monitoring methodology/08/.											
	Monitoring equipment	The Main Energy meter and Back up Meter of accuracy class 0.5s have been used/16/. (Calibration details of meter is provided separately in this section, under the heading "Calibration of meters"). Details of the main and back up meter are tabulated below: <table border="1"> <tr> <td></td><td>Main Meter</td><td>Spare Meter</td><td>Main Meter</td><td>Spare Meter</td></tr> <tr> <td></td><td>(current)</td><td>(current)</td><td>(removed on</td><td>(removed on</td></tr> </table>				Main Meter	Spare Meter	Main Meter	Spare Meter		(current)	(current)	(removed on
	Main Meter	Spare Meter	Main Meter	Spare Meter									
	(current)	(current)	(removed on	(removed on									

			14/10/2020)	14/10/2020)
Brand	EMH	EMH	ACTARIS	ACTARIS
Type	LXQJ-XC	LXQJ-XC	SL 7000 761 A	SL 7000 761 A
Class	1S	1S	0.5 S	0.5 S
Serial No	9420180	9420181	65000694	65000695
Installation Date	14/10/2020	14/10/2020	13/12/2010	13/12/2010
Test Dates	23/07/2020	23/07/2020	18/07/2018 - 23/07/2020	18/07/2018 - 23/07/2020
The first pair of main (Serial No. 65000694) and spare meters (Serial No. 65000695) were installed on 13/12/2010 and changed on 14/10/2020 to the new pair of meters i.e. Main meter (9420180) and Spare meter (9420181). Since MP1 is from 21-February-2014 to 20-February-2020 the meters recording data during MP were the previous meters.				
Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as	Yes, accuracy class of meter i.e 0.5S is in line with registered monitoring plan/01/ and as per Provisional Acceptance Document /25/. This has been verified via the first index protocol/39/, meter technical specifications/16/ and the calibration certificates/12,39/. As per Article 2 of the Communiqué of Meters in Electricity Sector,/23/ the meters to be used in the electricity market shall be compliant with the standards of Turkish Standards Institute or IEC and have obtained "Type and System Approval" certificate from the Ministry of Trade and Industry.' Therefore, Ministry of Trade and Industry (Ministry) is responsible from control and calibration of the meters. Also, according to Article 11 of this Communiqué, The PA must have IEC-EN 60687 meters of class of 0.5s,			

	per the manufacturer's specification?	which means error interval for measuring is in +/-0.5% range which is well acceptable according to rules. Paragraph b) of the Article 9 of the 'Regulation of Metering and Testing of Metering Systems' (Regulation)/23/ of Ministry states that: 'b) Periodic tests of meters of electricity, water, coal gas, natural gas and current and voltage transformers are done every 10 years.' Therefore, periodic calibration of the meters will be done every 10 years according to the regulation. The calibration of the meters will be done by the Ministry of Trade and Industry. The main meter and backup meter were calibrated upon installation as per the First index protocol certificates/39/. This was followed by standard meter tests in July 2018 and July 2020, as confirmed from meter test report /13/. The meter was replaced on 14/10/2020 as confirmed from the 'Meter and Measurement Transformers Change Protocol" They are next scheduled to be calibrated in 2030.
	Calibration frequency /interval:	The calibration frequency of the meters as per Turkish regulation is confirmed to be once in 10 years (Paragraph b) of the Article 9 of the 'Regulation of Metering and Testing of Metering Systems' (Regulation)/2/ of Ministry). The calibration of the meters will be done by the Ministry of Trade and Industry. The main meter and spare meter (old) were calibrated upon installation as per the First index protocol certificates/39/. As per Paragraph b) of the Article 9 of the 'Regulation of Metering and Testing of Metering Systems' (Regulation)/23/ of Ministry states that: 'b) Periodic tests of meters of electricity, water, coal gas, natural gas and current and voltage transformers are done every 10 years.' Therefore, periodic calibration of the meters will be done every 10 years according to the regulation. The calibration of the meters is done by the Ministry of Trade and Industry. Since the meters were installed on 13/12/2010 and replaced on 14/10/2020 i.e. before 10 years were up, hence, no calibration was required during the current monitoring period.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring	As per the monitoring plan in the registered PD/1/, meter is supposed to be calibrated once in 10 years. The main meter and backup meter (old) were calibrated upon installation as per the First index protocol certificates/39/. The main meter and backup meter were calibrated upon installation as per the First index protocol certificates/39/.

plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, or as per the manufacturer's specifications?	This was followed by standard meter tests in July 2018 and July 2020, as confirmed from meter test report /13/. The meter was replaced on 14/10/2020 as confirmed from the 'Meter and Measurement Transformers Change Protocol' They are next scheduled to be calibrated in 2030.
Is the calibration of measuring equipment carried out by an accredited person or institution?	Yes, the calibration is conducted by Ministry of Trade and Industry (Türkiye)/12,39/.
Is(are) calibration(s) valid for the whole reporting period?	Yes, it is valid for entire Monitoring Period. The meters were calibrated upon installation on 13/12/2010, which holds merit for 10 years as per Paragraph b) of the Article 9 of the 'Regulation of Metering and Testing of Metering Systems' (Regulation)/23/ of Ministry. Given that monitoring period is from 21-February-2014 to 20-February-2020, i.e. with the 10 year mark from date of calibration, the calibration is valid for the entire monitoring period.
Is the calibration carried out for a measuring range comparable with the range for which measurements have been carried out?	Yes, the calibration has been carried out for a measuring range comparable with the range for which measurements have been carried out. Since the meters were replaced before 10 years were up from date of installation of details, hence as per the 'Regulation of Metering and Testing of Metering Systems' no further calibration would be required for the meters, apart from that done during commissioning as described above.
How were the values in the monitoring report verified? If applicable, has the reported data been cross-checked with other available data	Monthly values for entire monitoring period are reported in the monitoring report/02/ and the ER calculation sheet/04/. Cumulative value of $EG_{Facility,y}$ for entire monitoring period is reported in the monitoring report, however the monthly values are reported in the ER calculation sheet/3/. The monthly values were verified from monthly EPIAŞ data/11/ is used as a main source. Cross-check is done by using the invoices. Value of this parameter for the current monitoring period was verified as net 210,015.91 MWh/4/.

	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the adequate QA/QC procedures were followed by the PP. The data transfer process for the said parameter is in accordance with the registered monitoring plan/1/ and applied methodology/8/. Yes, all the stakeholders, namely, the National Grid Authority, the PP and the HEPP Chief/28/ for the project, implemented the adequate QA/QC procedures. The data transfer process for the said parameter is as follows: The meter reading is taken electronically by meter of TEIAS and archived electronically. This is reflected in the monthly reports which records day, peak and night hour electricity generation of the plant. Transmission losses are already deduced from the gross electricity generation, therefore, no other action is required. The meter reading is cumulative for UMUT 1, 2 and 3, hence it is multiplied with the constant 0.5787, the capacity of UMUT 1, UMUT 2 and UMUT 3 has been confirmed from the generation license. The VVB team has also assessed the Calculation pf constant factor provided in the ER Sheet and found it be appropriate.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	Not Applicable
Findings	No findings were raised	
Conclusion	The parameter has been monitored appropriately, in accordance with the registered monitoring plan (as per measurement methods and procedures to be applied) and applied methodology. The monitoring results were recorded consistently as per the approved frequency in the monitoring plan.	

GHG Calculations:

The proposed project activity has applied baseline methodology as mentioned in the large-scale methodology ACM0002 version 12.1.0, "Grid-connected electricity generation from renewable sources". Accordance with the applied methodology emission reductions is calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

LE_y = Leakage emission in year y (t CO₂e/yr)

Calculations of Baseline emissions:

As per the applied methodology baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected hydro power plant.

The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

where,

BE_y = Baseline emissions in year y (tCO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

EF_{grid,CM,y} = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

As per the paragraph 43, the calculation of EGPJ,y is different for Greenfield plants, capacity additions, retrofits, rehabilitations, and replacements. Since the proposed project activity is Greenfield plant, hence as per the paragraph 44 of ACM0002, Version12.1.0,

$$EGPJ,y = EG_{facility,y}$$

Where,

EGPJ,y = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

EG_{facility,y} = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$\text{Hence, } BE_y = EG_{facility,y} * EF_{Grid,CM,y}$$

$$\begin{aligned} BE_y &= 210,015.91 \text{ MWh} \times 0.551 \text{ tCO}_2/\text{MWh} \\ &= 115,715 \text{ tCO}_2\text{e} \end{aligned}$$

Calculation of project emission

Project emissions are calculated as follows:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

where

PE_y= project emissions in year y (tCO₂e/yr)

PE_{FF,y}= project emissions from fossil fuel consumption in year y (tCO₂/yr)

PE_{GP,y}= Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y

PE_{HP,y}= project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

PE_{FF,y} is zero as there will be no fossil fuel consumption to generate electricity and PE_{GP,y} is zero as the project is not a geothermal project activity.

In order to calculate project emissions from water reservoir of the plant, power density should be calculated. The power density of the project activity (PD) is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

where

PD= Power density of the project activity

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity
 Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W).

For the new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity when reservoir is full (m²)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero.

As the project activity is not extension of another project, Cap_{BL} and A_{BL} are zero, then

$$PD = \frac{Cap_{PJ}}{A_{PJ}}$$

As in the case of UMUT II HEPP the power density is calculated as:

$$PD = 24,450,000/113,696 = 215 \text{ W/m}^2$$

As per the applied methodology ACM0002 v12.1.0 , for most renewable power generation project activities, PE_y = 0. Since the project activity is a hydro energy-based power generation and the power density is more than 10 W/m² in the plant, the project emissions from reservoir have been considered to be zero.

Hence, PE_y = PE_{HP,y} = 0 where PE_y= project emissions in year y (tCO₂e/yr) PE_{HP,y}= project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

Calculation of leakage emissions:

As per the applied methodology ACM0002 v12.1.0, no leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, and transport). These emissions sources are neglected.

Therefore, LE_y = 0.

Calculation of Net emissions reductions:

Emission reductions related to project activities must be estimated using ACM0002's formula 11:

$$ER_y = BE_y - PE_y - LE_y$$

where

ER_y= emission reductions in year y (tCO₂/yr)

BE_y= baseline emissions in year y (tCO₂/yr)

PE_y= project emissions in year y (tCO₂/yr)

LE_y= Leakage emissions in year y (tCO₂/yr)

Since PE_y=0 and LE_y =0

The emission reduction as per the applied methodology equals the baseline emissions (project emissions and leakage emissions for such project activities is considered zero).

$$ER_y = BE_y$$

The formula provided for the calculation of baseline emissions is:

Thus, BE_y = EGPJ, y * EF grid, CM, y

Thus, BE_y = 210,015.91 MWh X 0.551 tCO₂/MWh

$$ER_y = BE_y = 115,715 \text{ tCO}_2\text{e}$$

The total emission reduction during the current monitoring period – 21/02/2014 to 20/02/2020 is 115,715 tCO₂e.

The assessment team confirms that the appropriate methods and formulae for calculating GHG emission have been followed. The assumptions, emission factor and default values that were applied in the calculations are justified. The actual emission reduction achieved during the current monitoring period is higher than the estimated emission reduction as described in the registered VCS PD/1/.

All the data were made available and have been monitored as per the required monitoring frequency.

No sampling has been done by the VVB as 100% of the data were verified for current monitoring period.

4.6 Non-Permanence Risk Analysis

Not applicable

5 VERIFICATION OPINION

Earthood Services Private Limited (Earthood) was contracted by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. to perform the independent verification of the emission reductions for the VCS project activity (VCS ID – 1011) “UMUT II Hydropower Plant, Türkiye” for the monitoring period 21/02/-2014 to 20/02/2020 as reported in the Monitoring report version 5.0, dated: 22/02/2024/2/.

Earthood commenced the verification based on the baseline and approved methodology ACM0002, Version 12.1.0 “Grid connected electricity generation from renewable sources”/8/, the monitoring plan contained in the VCS PD/1/, VCS guidelines/6/7/, Monitoring Report/2/ as per the process described under section 2 of this report.

Earthood verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the verification by obtaining evidence and other information and explanation that Earthood considered necessary to give reasonable assurance that the reported GHG emission reductions are fairly stated.

In our opinion, the GHG emission reductions for the project activity for the monitoring period 21/02/-2014 to 20/02/2020 (both days included) are fairly stated in the monitoring report/2/. The GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002/8/ and the VCS standard/7/.

Verification period: From 21/02/-2014 to 20/02/2020 (both days included).

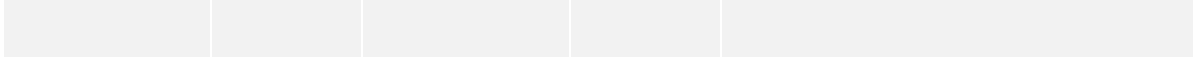
Verified GHG emission reductions and removals in the above verification period, broken down by calendar year:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2014 (21/02/2014 – 31/12/2014)	12,637	0	0	12,637
2015	19,937	0	0	19,937
2016	22,904	0	0	22,904
2017	20,266	0	0	20,266
2018	17,187	0	0	17,187
2019	21,372	0	0	21,372
2020 (01/01/2020 – 20/02/2020)	1,412	0	0	1,412

Total	115,715	0	0	115,715
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The difference between the Ex-ante emission reductions/removals and the achieved emission reductions has been calculated in the ER calculation sheet/4/ as -49%. The justification for the decrease in achieved emission reductions has been mentioned in the table below:

Year	Ex-ante emissions reduction s/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
21/02/2014-20/02/2020	227,254	115,715	-49%	The estimated electricity generation value is taken from the Generation Licence. However, the actual generation values are lower than the estimated. This is due to water flow rate affecting the electricity generation. The difference is caused by the water flow that the power plant received. Since, the power generation is aligned with the water flow. The project activity received less water flow and due to that the electricity generation was recorded low for the current monitoring period. Minimum water release is determined by DSI (State Hydraulic Works) distinctly for each hydro power plant. There is no set limit as “x m3/s” but DSI monitors these flows and provides their measurements to PP (provided for evidence). There have not been any disputes between DSI and the Umut HEPP. The “water usage agreement”/37/ and “Lifewater records (DSI approved)”/38/ have been assessed to corroborate the same.



Approved by



Dr. Kaviraj Singh

Managing Director

Earthood Services Privated Limited

Date: 08-08-2024

Place: Gurugram, Haryana

APPENDIX I: DOCUMENTS REVIEWED

	Author	Title	Reference to documents	Provider
1.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Project Description	Version 1.2, dated 23/09/2011	PP
2.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Monitoring Report	Version 1.0	PP
3.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Final Monitoring Report	Version 0.6, dated 08/08/2024	PP
4.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	ER Spreadsheet	Corresponding to the MR	PP
5.	TÜV Rheinland Energie und Umwelt GmbH	Validation Report	Version 1.0, dated 18/08/2011	Others
6.	VCS	VCS Program Guide	Version 4.4, dated 29/08/2023	Others
7.	VCS	VCS Standard	Version 4.5, dated 04/10/2023	Others
8.	CDM	ACM0002, Version 12.1 “Grid connected electricity generation from renewable sources”	Version 12.1	Others

9.	CDM	Tool for the demonstration and assessment of additionality	Version 07.0	Others
10.	CDM	Tool 7: Tool to calculate the emission factor for an electricity system	Version 02.1.0	Others
11.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	EPIAS Screenshots	For the current monitoring period: 04-October-2012 to 03-October-2018	PP
12.	TEIAS	Meter and Measurement Transformers Change Protocol	2020	PP
13.	TEIAS	Meter Test Reports for main (Serial No. 65000694) and spare meters (Serial No. 65000695)	18/07/2020 23/07/2020	PP
14.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Grievance register	21-February-2014 to 20- February -2020	PP
15.	VVB	Photographs of the project site and energy meters	-03/10/2023	PP
16.	Manufacturer's protocol	Technical specifications of the equipments	-	PP
17.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Declaration for no participation under any other programs other than VCS	15/12/2023	PP
18.	TEIAS	Meter reading reports	21-February-2014 to 20- February -2020	PP
19.	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş	Maintenance details	For the current monitoring period: 21-February-2014 to 20- February -2020	PP

20.	VCS	VCS Verification Report Template	Version 4.2 21/12/2022	Others
21.	VCS	VCS Monitoring Report Template	Version 4.2 21/12/2022	Others
22.	PP	Local Stakeholder Consultation evidence LSC Report	10/04/2009	PP
23.	Government of Turkiye	Regulation on Metering and Metering Devices- • Communiqué for Metering Devices to be used in the Electricity Market • 'Regulation of Metering and Testing of Metering Systems' (Regulation)	-	PP
24.	Ministry of Labour and Social Security	Employment Records – Social Security details of employees		PP
25.	T.C. Enerji Ve Tabii Kaynaklar Bakanligi	Provisional Acceptance Document (proof of commissioning)	Project Approval Date and Number: 21/02/2014; 13845	PP
26.	Government of Turkiye	Host Country Criteria (Turkiye)	-	Others
27.	VCS	Project Webpage https://registry.terra.org/app/projectDetail/VCS/1012	Last Accessed on 19/02/2024	-
28.	VVB	On Site Observation Records, physical interviews	03/10/2023	
29.	T.C Enerji Piyasasi Duzenleme Kurumu	Generation License (License Number: EU/1590-1/1156 Amendments 8	01/05/2008 24/01/2014 03/09/2021	PP

		Amendment 14		
30.	PP	Copy of Communication to Verra regarding Change in PA License Holder	17/01/2022	
31.	Google Earth	.KML file of PA location Longitude:37.026944 Latitude: 40.713333	Last accessed on 07-08-2024	PP
32.	Government of Türkiye	Government of Türkiye's Official Gazette No:26939 From the Ministry of Environment and Forestry: BY-LAW ON ENVIRONMENTAL IMPACT ASSESSMENT (EIA)	July 17th 2008, THURSDAY	Others
33.	Ministry of Environment and Forestry General Directorate of Environmental Impact, Assessment and Planning	EIA Positive Document – UMUT 1-3	17/12/2009	PP
34.	RHG Enertuk and TEIAS	System Usage Agreement	24/11/2021	PP
35.	CINAR Engineereing	EIA Report (UMUT 1-3)	September 2009	PP
36.	UNFCC	2006 IPCC Guidelines for National Greenhouse Gas Inventories	Volume 2; 2006	Others
37.	General Directorate of State Hydraulic Works and Nisan Electromekanik Enerji Sanayi ve Ticaret K	Agreement on Water Use Rights and Operational Principles Of Umut Regulator and Hydroelectric Energy Production Facility	15/02/2022	PP

38.	Boydak Energy Inc.	Lifewater Records UMUT 2 HES Cansuyu Agy Data and Analysis UMUT 2 HES DDERE-1REG-Cansuyu AGy Data and Analysis UMUT 2 HES DDERE-2REG-Cansuyu AGy Data and Analysis UMUT 2 HES MySTAN REG-Cansuyu AGy Data and Analysis UMUT 2 HES YALAMA REG-Cansuyu AGy Data and Analysis)	2018- 2019	PP
39.	TEIAS	First Index Protocol for Main meter (Serial Number 65000694) and Spare Meter (65000695)	14/10/2022	PP
40.	Ordy Metropolitan MUnicipality	Waste Records- Certificate of Receipt	For the current monitoring period: 04-October-2012 to 03-October-2018	PP

APPENDIX II: ABBREVIATIONS

Abbreviations	Full texts
ABT	Availability Based Tariff
BEF	Baseline Emission Factor
BM	Build Margin
CAR	Corrective Action Request
TEIAS	Türkiye Elektrik İletim A. Ş. (Turkish Electricity Transmission Company)
EPIAS	Enerji Piyasaları İşletme A.Ş. (Energy Exchange Istanbul)
CL	Clarification Request
CMS	Central Monitoring System
CMP	Conference of Parties Serving as Meeting of Parties
CO2	Carbon dioxide
EB	Executive Board
FAR	Forward Action Request
GHG	Green House Gas
ISO	International Standards Organization
Kw	Kilowatt
kWh	Kilowatt hour
MFR	Multi-Function Relay
MR	Monitoring Report
MW	Megawatt
MWh	Megawatt-hour
O&M	Operation and Maintenance
PD	Project Description
PLF	Plant Load Factor

PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance and Quality Control
TOD	Time of the Day
UNFCCC	United Nations Framework Convention on Climate Change
HEPP	Hydro Electric Power Plant
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCS PD	VCS Project Description
VCUs	Verified Carbon Units

APPENDIX III: FINDINGS OVERVIEW

Table 1. CL from this verification

CL ID	01	Section no.	1.1 and 1.6 of the MR	Date:17/11/2023
Description of CL				
In the Monitoring report following discrepancies pertaining to the duration of the project activity were observed: As per the VCS registry (https://registry.terra.org/app/projectDetail/VCS/1012), it is mentioned that the project's first crediting period is from 01/10/2012 - 30/09/2022. However, Section 1.6 of MR states first crediting period to be from 21/02/2014 – 20/02/2024. Further the Project's first monitoring period is described in MR as 21-February-2014 to 20-February-2020 PP shall clarify the inconsistency and provide supportives for the same.				
Project proponent response				Date:14/12/2023
<i>The link that was provided is for Umut-I.</i> <i>In the PDD (provided), there is a sentence "... and the generation is expected to start on 01.10.2012.". As per the commissioning document dated as 21/02/2014, the start date of the project is taken as 21/02/2014. The commissioning was after the registration. The evidence document has been provided.</i> <i>So, the monitoring period is between 21-February-2014 to 20-February-2020.</i> <i>The dates in the registry system will be corrected after the submission.</i>				
Documentation provided by project proponent				
VVB assessment				Date : 13/02/2024
The Monitoring Report has been revised and states monitoring period to be from 21/02/2014 20/02/2020 which is found to be consistent throughout the updated MR and with the ER Sheet. CLOSED				

CL ID	02	Section no.	ER Sheet	Date:17/11/2023
Description of CL				
PP shall clarify the following discrepancies arising in the ER Calculation Sheet:				

1. In tab 'Detail', the combined gross (Cell Q47) and Net (Cell S47) Electricity generation as per meter records for the month of September for the year 2017 is mentioned as 108.680 MWh and 76.34 MWh respectively. However, as per EPIAS records, the combined gross (Cell T47) and Net (Cell V47) Electricity generation as per meter records for the month of September for the year 2017 is given as 161.55 MWh and 114.08 MWh respectively. PP is requested to clarify the significant increase in energy generation recorded as per the EPIAS records in comparison to the value of energy generation as per the meter records for the specified duration of the monitoring period. 2. Further PP shall justify the significant difference for Net electricity generated between readings from 'Meter Records '(Column I) and 'EPIAS Records'(Column L), as reported under Column 'O' titled 'Difference Analysis' whith the variance being as large as 1150.664% (Cell O46). PP is requested to clarify the inconsistency between the reported values.	
Project proponent response	Date: 14/02/2023
1. <i>There are differences in every month as can be seen in the O column, the difference analysis. These differences may be caused by that the data of EPIAS and TEIAS are for whole Umut HEPP (meaning Umut-I+Umut-II+UmutIII). The portion of the installed capacity of Umut-II differs from Umut-I and Umut-III.</i> 2. <i>There was a slide in the values, and now it has been corrected. The actual "Difference Analysis" is not as high as mentioned.</i>	
Documentation provided by project proponent	
VVB assessment	Date : 13/02/2024
1. <i>VVB has analysed the difference in EPIAS and TEIAS records and found the differences to be comparable in nature across months, hence found appropriate.</i> 2. <i>The calculations have been revised and the ER sheet has been updated accordingly. All outlying values have been rectified ,</i> <i>The finding stands closed.</i>	

CL ID	03	Section no.	Section 1.1 of the MR	Date : 26/11/2023
Description of CL				
Under Section 1.1 of the Monitoring report, it is stated that "RHG Enertürk Enerji Üretim ve Ticaret A.Ş. operates 'Umut-II Hydropower Plant, Turkey Project' with 24.450 MWe (25.305 MWm) run-of-river hydropower electricity plant (HEPP) is located on Karakus River, Ordu, Türkiye." According to which the HEPP plant is located on the Karakus River. However as per the Project Design Description, and Validation Report ver 1.0 available on the VCS Project webpage, The project activity is located on the Gokcebayir River. PP shall clarify if the above two refer to the same water body as the names can be used interchangeably, and make the necessary corrections in the Monitoring report as appropriate.				

Project proponent response	Date : 14/12/2023
The Gökçebayır River is mentioned as “Karakuş River” by the villagers, and by plant managers during the site visit. The contradiction has been corrected throughout the MR.	
Documentation provided by project proponent	
VVB assessment	Date : 13/02/2024
The MR has been revised to mention ‘Karakus River’ in place of ‘Gökçebayır River’. This is found to be in line with the registered PDD and OSV observations.	
The finding is now Closed.	

CL ID	04	Section no.	On site Audit	Date : 20/11/2023
Description of CL				
1. During on site visit it was observed by the verification team that the HPP is not operational throughout the year and undergoes planned maintenance for a specified time period during the year. PP is requested to share maintenance records to substantiate the planned maintenance undertaken during the monitoring period. 2. It was observed during the site visit that Umut-II HPP has two DG sets installed. However, the total project emissions from burning diesel have been indicated as zero in cell M91 of the worksheet 'Detail' in the ER Calculation sheet. PP is requested to clarify the operability of the DG sets at the HPP and provide supporting documents to substantiate their operability throughout the monitoring period.				
Project proponent response				Date : 14/12/2023
1. The maintenance records have been provided. 2. The operability about diesel generators and supportive documents have been provided.				
Documentation provided by project proponent				
VVB assessment				Date: 24/01/2024
The evidences provided have been found appropriate and in line with on site observations made by the VVB team.				
The finding stands CLOSED.				

CL ID	05	Section no.	1.11 of the MR	Date : 20/11/2023
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Description of CL	
In section 1.11 of the Monitoring report, titled “Sustainable Development Contributions”, under table 5, column ‘Current Project Contributions’ for SDG target 8.5, it is mentioned that “<i>Due to the activity of the power plant, 27 people were employed.</i>”. However, the employment records provided by the PP were checked for both Umut-I HPP and Umut-II HPP by the verification team and it was found that the people employed by Umut-I HPP are same as the people employed by Umut-II HPP. PP shall clarify the exact number of people that were employed by Umut-II HPP during the current monitoring period and provide a supportive accordingly.	
Project proponent response	Date : 14/12/2023
<u>Response-1:</u> <i>A supportive document has been provided, and the number has been corrected.</i> <i>There are 12 employees working for Umut-II, of who, 2 are common for Umut-I, Umut-II, and Umut-III HEPP. It is indicated in the MR. Evidence documents have been provided.</i>	
Documentation provided by project proponent	
VVB assessment	Date: 13/02/2023
The MR has been updated to reflect the actual number of people employed by UMUT 2. The finding stands closed.	

Table 3. CAR from this verification

CAR ID	01	Section no.	Section 4 of the MR	Date: 03/11/2023
Description of CAR				
Under section 4 ‘Data and Parameters’ of the MR the following have been observed: 1. The row for ‘Purpose of Data’ , as instructed in ‘vcs-monitoring-template-ver 4.2’ is not included for any ex ante or monitored parameter. PP shall make the necessary inclusions. 2. The value of Parameter EGpj,facility,y in MR, Section 4.2 is inconsistent with the ER Calculation sheet , Tab 'Details', Cell 'H80'. The value described in MR is same as that in PDD ver1.2, however being a monitored parameter, this similarity is highly unlikely. PP to address the discrepancy. 3. In section 4.2, for parameter ‘EGPJ,facility,y’, PP shall mention the due date of calibration for the monitoring equipment i.e. main (serial no. 9420180) and spare (serial no. 9420181) meters, under QA/QC procedures.				

Project proponent response	Date: 14/12/2023
1. The purpose of data rows have been included. 2. The value of the net generation is “210,015.91 MWh” in the Excel cell H80, the value in the MR is the same. The PDD version that has been used is 1.4. There is no discrepancy. 3. The calibration due date has been mentioned.	
Documentation provided by project proponent	
VVB assessment	Date: 15/02/2024
1. The MR has been updated and now includes ‘purpose of data’ for all parameters, ‘ex-ante’ and ‘monitored’. 2. The value of parameter ‘EGpj,facility,y’ has been assessed and found to be consistent between ER Sheet and MR. 3. The MR has been updated to reflect the calibration dates.	
Finding stands CLOSED.	

CAR ID	02	Section no.	Section 1.10 of MR	Date: 17/11/2023
Description of CAR				
According to the guidelines under Section 1.10 of VCS-Monitoring report template v4.2, which state that “Include the following information, as applicable: • <i>Emission Trading Programs and Other Binding Limits: Where the project reduces GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading (as identified in the project description, or where such programs or mechanisms have subsequently emerged) demonstrate that net GHG emission reductions or removals generated during this monitoring period have not been used for compliance under such programs or mechanisms. Examples of appropriate evidence are provided in the VCS Standard.</i> • <i>Other Forms of Environmental Credit: Indicate whether the project has sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period. Include all relevant information about the GHG-related environmental credits and the related program. Additionally, provide a list of all and any other programs under which the project is eligible to create another form of GHG-related environment credit.</i> • <i>Supply Chain (Scope 3) Emissions: A supply chain is a network of organizations (e.g., manufacturers, wholesalers, distributors, and retailers) involved in producing, delivering, and selling a product or service to the consumer. Scope 3 inventory emissions are all indirect upstream and downstream GHG emissions in an organization’s supply chain. Carbon project activities may impact the emissions of goods and services in a supply chain and, therefore, Scope 3 emissions. If a project affects emissions associated with a good or service, demonstrate that a public statement(s) by the owner(s) or retailer(s) of the impacted good(s) or service(s) or project proponent (as applicable) has been made throughout the project crediting period. Where applicable, also demonstrate that the impacted good or service's producer(s) or retailer(s) have been notified of the project and the potential risk of Scope 3 emissions double claiming via email. Evidence of the public statement(s) and email(s) must be provided in this report or attached as an appendix.”</i> PP is requested to kindly review and revise the section and demonstrate about the supply chain (scope 3) emissions as per the template guidelines.				

Furthermore, PP shall provide supplement 'Declaration stating that the VCU generated by the project activity will not be used in any other carbon offsetting program', as supportive for claims made under Section 1.9 and 1.10 of MR.	
Project proponent response	Date: 14/12/2023
<u>Response-1:</u> <i>The section has been revised and the Declaration Letter has been provided.</i>	
Documentation provided by project proponent	
VVB assessment	Date: 15/02/2024
PP has updated section 1.10 wrt 'Supply Chain (Scope 3) emissions', and adequately explains scope 3 emissions specific to the PA.	
The finding stands Closed.	

CAR ID	03	Section no.	ER Sheet	Date: 17/11/2023
Description of CAR				
In the ER Calculation Sheet, - The electricity generation values as per the EPIAS records mentioned in Cell T53 and U53 for the month of November 2016 have been found missing. PP shall review and revise the worksheet to include the energy generation values for the specified duration. - It has been observed that the Gross energy generation values as per the meter records given in worksheet titled "Detail", given in column C for Umut-II HPP have been sourced incorrectly. For example, the gross generation value for the month of February 2014 mentioned in cell C4 of the worksheet "Tab" has been sourced from cell Q5 which signifies combined (Umut-I, Umut-II and Umut-III HPP) gross generation for the month of February 2014. PP shall update the source of the values appropriately in accordance with the specified months.				
Project proponent response				Date: 14/12/2023
- The mentioned values is included in the ER sheet. - The mistake has been corrected.				
Documentation provided by project proponent				
VVB assessment				Date: 15/02/2024
The ER has been updated to reflect the necessary changes. The finding stands CLOSED.				

Table 4. FAR from this verification

FAR ID	01	Section No.	Start Date	Date: 15/02/2024
Description of FAR				
Change in the start date of the PA has resulted in change in the duration of the first and second crediting period. The first crediting period now is from 04/10/2012 – 03/10/2022, and the second crediting period is for a subsequent 10 years. However the VCS registry states the second monitoring period to be from 01/12/2021 to 30/11/203, which overlaps with the first crediting period. The subsequent VVB shall ensure that post issuance for 1st monitoring period, the registry webpage is updated to reflect the correct duration of the second crediting period, in line with the altered start date of PA.'				
Project proponent response				Date: DD/MM/YYYY
-				
Documentation provided by project proponent				
-				
VVB assessment				Date: DD/MM/YYYY
-				

APPENDIX 4: COMPETENCY STATEMENTS

Competence Statement	
Name	Ranjan Singh
Education	BSc (Physics), MBA (Marketing)
Experience	13 Years
Field	Power, Utilities and Renewables
Approved Roles	
Team Leader	YES (VM only)
Validator	YES
Verifier	YES
Local expert	YES (India)
Financial Expert	NO
Technical Reviewer	NO

TA Expert (X.X)	YES (TA 1.2)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	29/06/2023
Approved by	Deepika Mahala (Technical Manager)	Date	29/06/2023

Competence Statement			
Name	Akanksha Sengupta		
Education	M.Sc Environmental Studies, University of Delhi B.Sc Zoology, Hans Raj College, DU		
Experience	1 year		
Field	Environment Science and Policy		
Approved Roles			
Team Leader	NO		
Validator	YES		
Verifier	YES		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Reviewed by	Shifali Guleria (Quality Manager)	Date	21/02/2024
Approved by	Deepika Mahala (Technical Manager)	Date	21/02/2024

Competence Statement	
Name	Shifali Guleria
Education	M.Sc. (Environmental Studies and Resource Management), TERI University
Experience	3+ year
Field	Climate Change
Approved Roles	
Team Leader	YES
Validator	YES
Verifier	YES
Methodology Expert	YES (AMS-I.A., AMS-II.G., AMS-II.E., AMS-III.A.V., AMS-I.D, ACM0002)
Local expert	YES
Financial Expert	NO
Technical Reviewer	YES
TA Expert	YES (1.2, 3.1)

Reviewed by	Deepika Mahala	Date	18/02/2022
Approved by	Ashok Gautam	Date	18/02/2022

Competence Statement			
Name	Shreya Garg		
Country	India		
Education	M.Sc. (Climate Science & Policy), TERI University		
Experience	9 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., AMS.III.BL, ACM0002, ACM0012		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.1, TA 1.2, TA 3.1, TA 13.1)		
Reviewed by	Shifali Guleria	Date	21/12/2022
Approved by	Deepika Mahala	Date	21/12/2022

Competence Statement			
Name	Abhishek Koul		
Education	Bachelor of Technology in Mechanical Engineering		
Experience	1 year		
Field	Climate Change		
Approved Roles			
Team Leader	Yes (VM only)		
Validator	Yes (VM only)		
Verifier	Yes (VM only)		
Methodology Expert	NO		
Local expert	Yes (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	TA 1.2		
Reviewed by	Shifali Guleria (Quality Manager)	Date	08/08/2023

Approved by	Deepika Mahala (Technical Manager)	Date	08/08/2023
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Competence Statement			
Name	Kubra Agriman		
Education	BS Environmental Engineering		
Experience	2 years		
Field	Environmental Engineering		
Approved Roles			
Team Leader	NO		
Validator	NO		
Verifier	NO		
Methodology Expert	NO		
Local expert	Yes (Turkiye)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Trainee	Yes		
Reviewed by	Shifali Guleria, Quality Manager	Date	27/12/2022
Approved by	Deepika Mahala, Technical Manager	Date	27/12/2022